

# What do four sensors miss?

Establishing the observability and reliability limits of low-cost plantar sensor arrays for wearable gait characterisation

CATEGORY Engineering & Technology

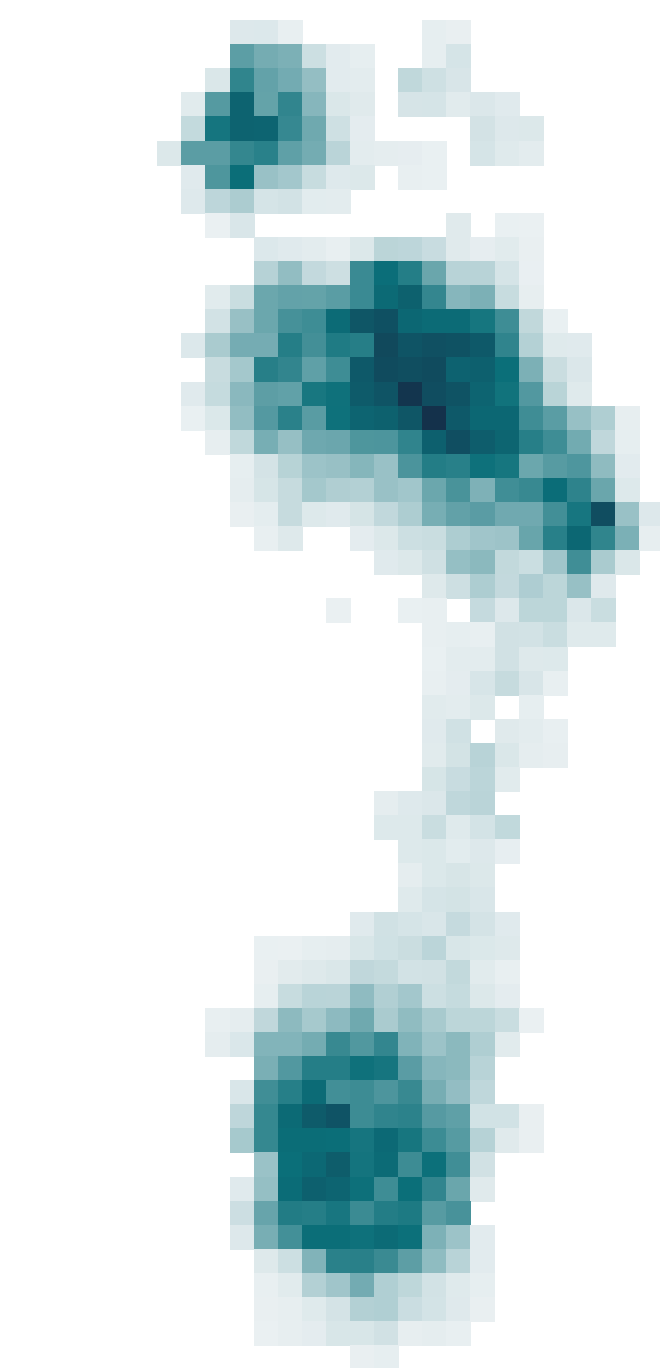
LEVEL UG

CODE NO. \_\_\_\_\_

## 01 / THE PROBLEM

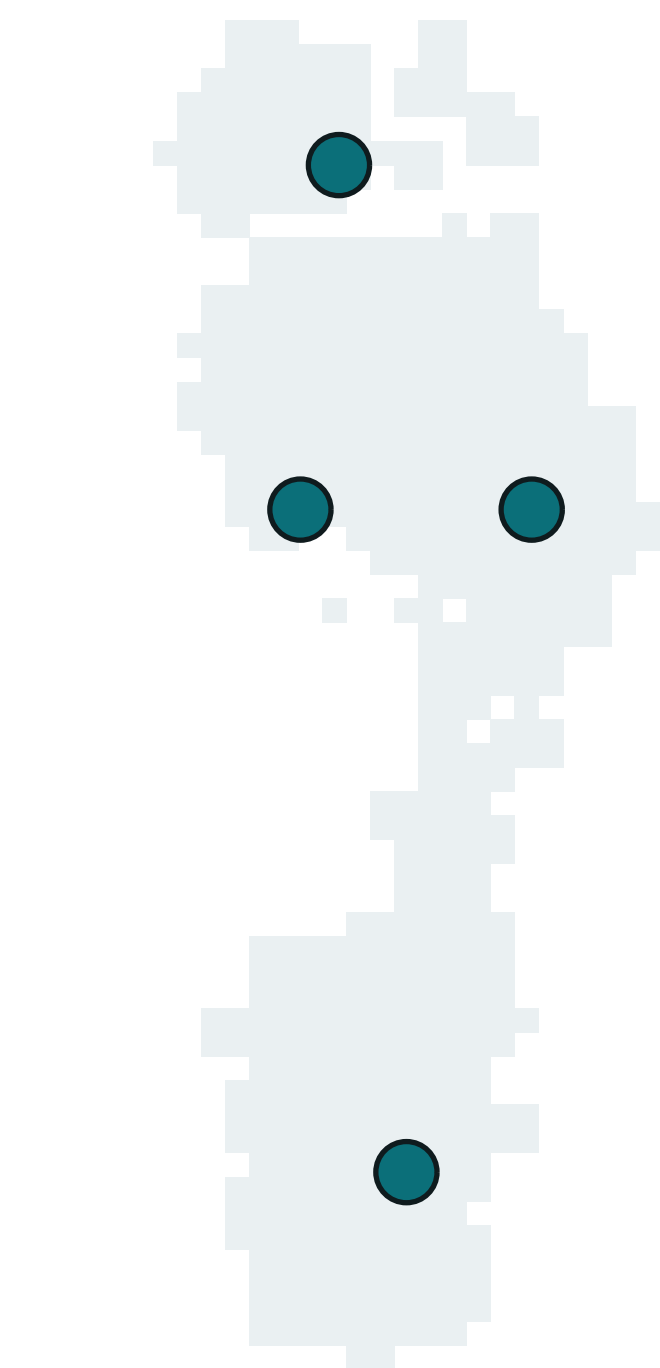
**A whole pressure field. Four small sampling areas.**

**Pressure map**



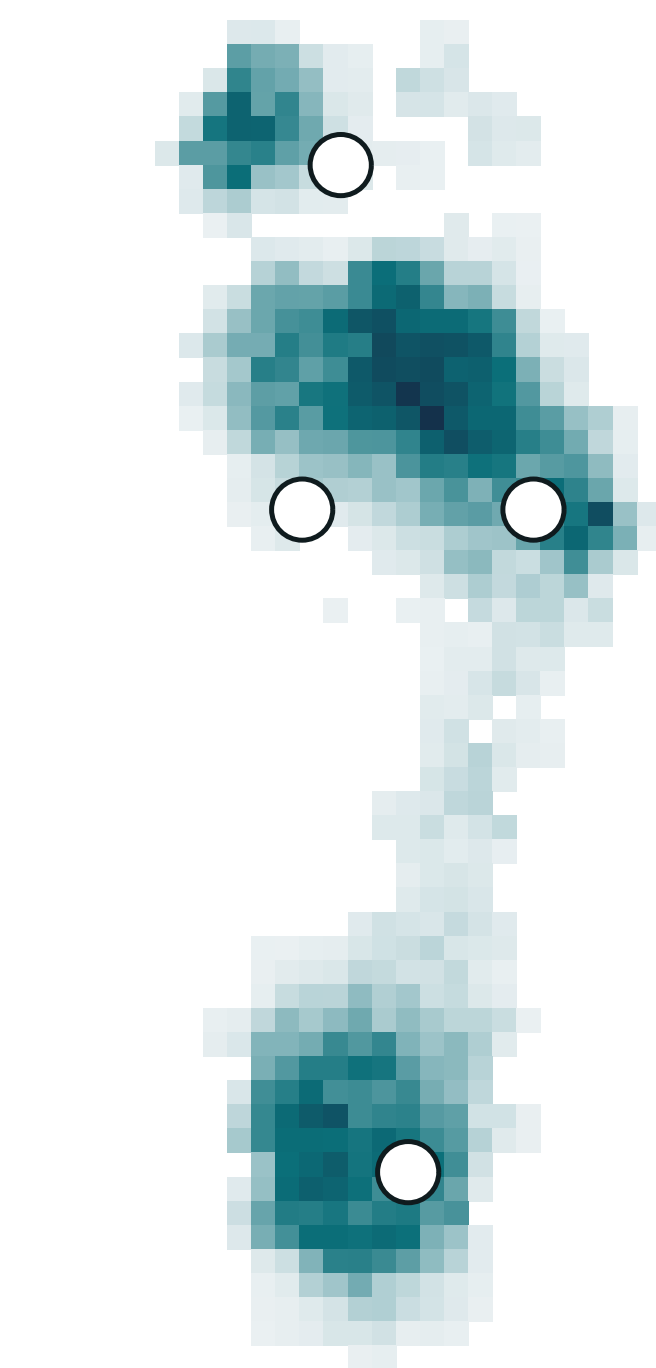
50 mm

**Sparse sensors**



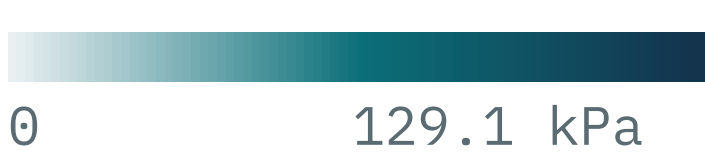
50 mm

**Unobserved area**



50 mm

**Dense reference**



Stance-mean pressure

**4 assumed discs**

Same right-foot example: participant 002, BF/W1, step 1.  
5 mm/cell, 12.7 mm discs. Toes up; medial side left.

**No direct measurement**

## R2 / SPATIAL COVERAGE

COMPLETED SUBSET ANALYSIS

**3.55%**

**Barefoot (BF)**

Range 2.76–4.25%

**2.18%**

**Sneaker (ST)**

Range 1.66–2.81%

Mean loaded area observed.  
Ranges show participant variation,  
not confidence intervals.

**The four-sensor model observes  
only a small fraction of loaded area.**

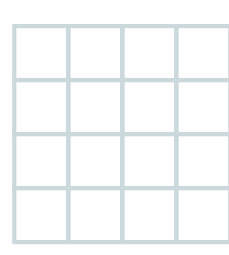
## 02 / HOW WE MEASURED IT



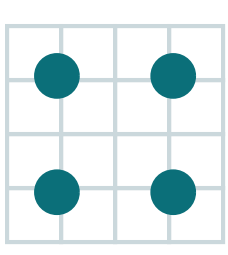
trials



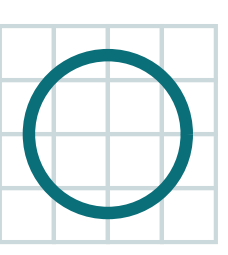
steps



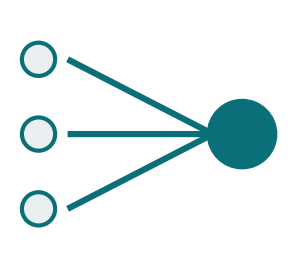
grids



masks



scores



means

**BF**  
**ST**

**Dataset**

StepUP-P150

**Step selection**

First eligible steps

**Pressure grid**

Side + pitch verified

**Sensor layout**

Fixed circular discs

**Overlap analysis**

Fractional cell area

**Aggregation**

Equal subject weight

**Results**

Conditions separate

**12**

**Participants**

IDs 002–013

**2**

**Conditions**

BF + ST, preferred speed

**576**

**Steps**

12 / foot / condition

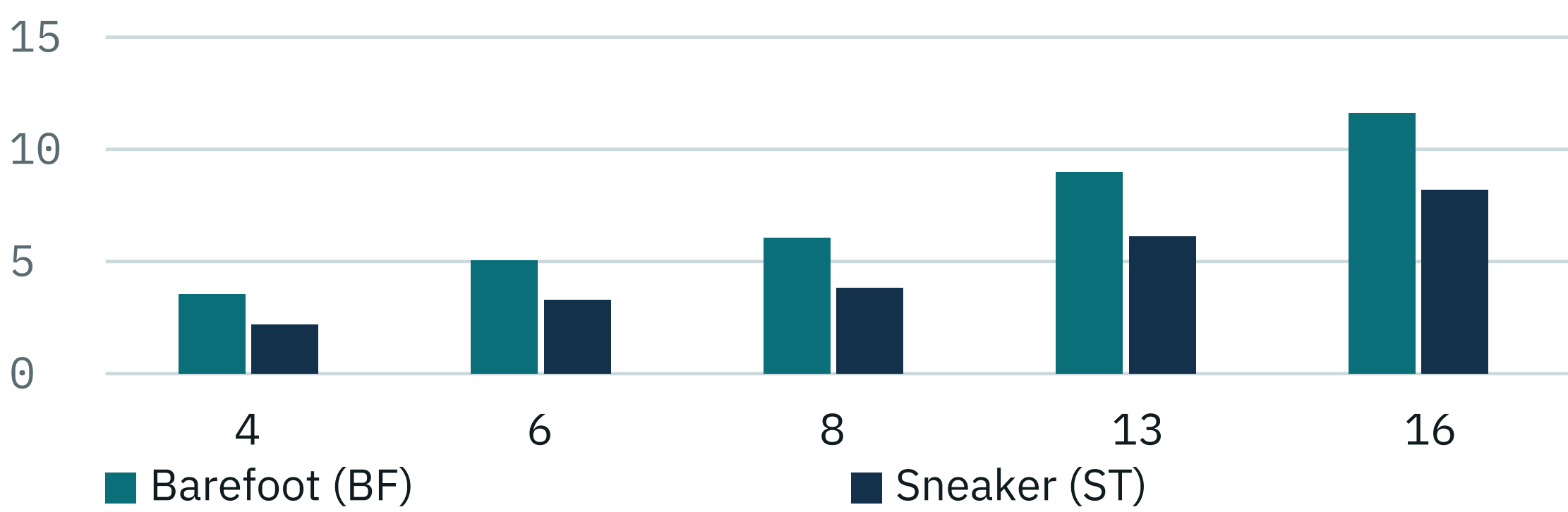
**5**

**Layout models**

4, 6, 8, 13, 16 sensors

## 03 / LAYOUT COMPARISON

Mean loaded area observed (%)

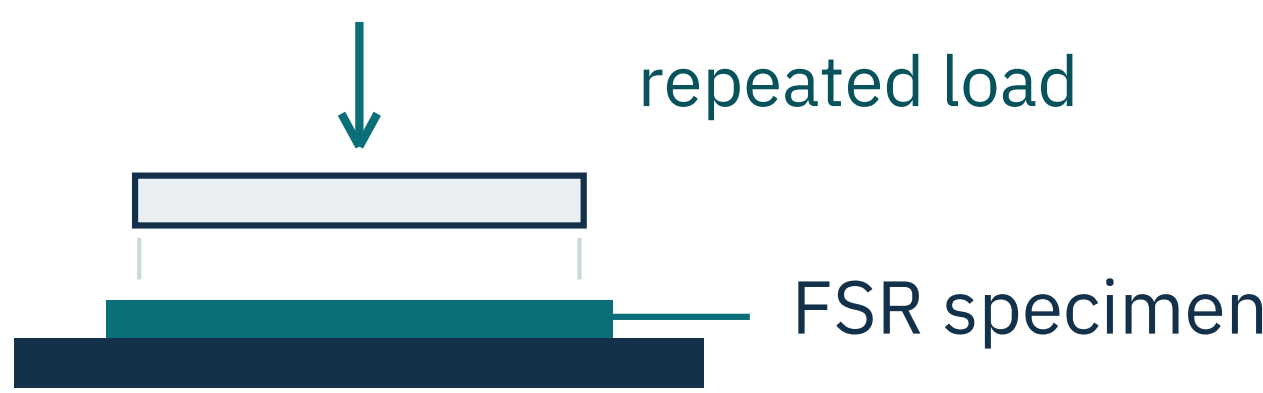


Sensor count in each layout model

**Count and position both change.**

## R1 / ENDURANCE

to be measured



Planned loading schematic

Compare calibration before and  
after repeated loading.

## R3 / MEASUREMENT VALIDITY

to be measured

**Drift checks**

Proposed gate

**VALID**  
Report

**DEGRADED**  
Suppress

**INVALID**  
Recalibrate

Test with aged sensors from R1.  
Firmware integration pending.

## CONCLUSION

Limited spatial coverage is the finding. Floor-pressure data do not validate in-shoe performance.  
Next: endurance testing, then validity-gate testing. Reliability remains to be measured.